## 1. Input / Output Functions in Python

1. Write a Python program that asks the user for their name and prints a greeting message.

2. Take two numbers as input from the user and print their sum.

3. Write a program that asks the user for their age and prints the age after 5 years.

4. Take a sentence as input and print it in uppercase.

5. Write a program that takes three numbers as input and prints their average.

6. Ask the user for a number and print its square.

7. Take a user's city name and print: "You live in ___".

8. Write a program that takes two numbers and prints their product.

9. Ask the user for a temperature in Celsius and print it.

10. Take two inputs (name and age) and display them in a sentence.

## 2. Variables

1. Create a variable storing your name and print it.

2. Store two numbers in variables and print their sum.

3. Swap the values of two variables.

4. Create variables for length and width and calculate the area of a rectangle.

~~5. Store your age in a variable and print it.~~

6. Create three variables and print them in one line.

7. Store a price in a variable and add tax to it.

8. Assign multiple values to multiple variables in one line.

9. Create variables for first name and last name and combine them.

10. Store a number and print its cube.

## 3. Type Casting

1. Convert an integer to a float.

2. Convert a float to an integer.

3. Convert a string `"123"` to an integer.

4. Convert an integer to a string and print it with text.

~~5. Take a number as input and convert it to an integer.~~

6. Convert a float to a string and print it.

7. Add two numbers taken as strings by converting them to integers.

8. Convert `"45.6"` to float.

9. Convert a number to string and concatenate it with another string.

10. Write a program to display the type of converted variable.

## 4. Operators

1. Write a program to add two numbers using arithmetic operators.

2. Write a program to check if one number is greater than another.

3. Use modulus operator to check if a number is even or odd.

4. Write a program using logical operators to check age eligibility for voting.

5. Demonstrate the use of assignment operators (`+=`, `-=`).

6. Write a program using comparison operators.

7. Calculate the power of a number using `**`.

8. Use floor division operator to divide two numbers.

9. Write a program that checks if two numbers are equal.

10. Combine arithmetic and comparison operators in one expression.

## 5. Expressions

1. Evaluate an expression involving addition and multiplication.

2. Write an expression to calculate the average of three numbers.

3. Create a Python expression to calculate the area of a circle.

4. Write an expression using parentheses to change precedence.

5. Evaluate the expression `10 + 5 * 2`.

6. Evaluate `(10 + 5) * 2`.

7. Write an expression that calculates simple interest.

8. Create a complex expression using multiple operators.

9. Write an expression to convert Celsius to Fahrenheit.

10. Evaluate an expression with variables.

## 6. String Formatting

1. Print a name and age using string formatting.

2. Format a string using `format()` method.

3. Use **f-strings** to print a sentence with variables.

4. Display a floating number with 2 decimal places.

5. Print a formatted price value.

6. Create a sentence using two variables with formatting.

7. Format a number with commas (100000 → 100,000).

8. Display a percentage using string formatting.

9. Align text using formatting.

10. Print a table row using formatted strings.